REVIEW + CLARIFICATIONS

Wakes:

- $$W \cong W_0 + x_w\left(W_x^D x_s - \frac{1}{2} W^Q x_w\right) + \\ + y_w\left(W_y^D y_s + W^Q y_w\right)$$

- $$W_\parallel \triangleq \frac{\partial}{\partial z} W(\vec{\rho_s}, \vec{\rho_w}, z) \cong W_0'(z)$$ ~ consideration only for the long. problem

- $$\vec{W_\perp} \triangleq \vec{\nabla}_\perp W = \left(\frac{\partial}{\partial x_w} W, \frac{\partial}{\partial y_w} W\right) = (w_x, w_y)$$

$$\Rightarrow \vec{W_\perp} \cong \left(W_x^D x_s - W^Q x_w, \; W_y^D y_s + W^Q y_w\right)$$

- P. W. $\rightarrow$ $$\frac{\partial}{\partial z}\vec{W_\perp} = \vec{\nabla}_\perp w_\parallel$$ here, $w_\parallel \cong W_0'(z)$ shouldn't be used, otherwise $\vec{W_\perp} = 0$ always

- Cylind. symmetry: $W_x^D = W_y^D = W^D$, $W^Q = 0$

$$\Rightarrow \vec{W_\perp} = W^D \cdot (x_s, y_s) = W^D \cdot \vec{\rho_s} = w_\perp \cdot \hat{\rho_s}$$

(#)

$$\frac{1}{c}\int_{-\infty}^{\infty} \frac{\partial W}{\partial z} e^{i\omega z/c} dz = \frac{1}{c} W e^{i\omega z/c}\Big|_{z=-\infty}^{z=\infty} - \frac{i\omega}{c^2}\int_{-\infty}^{\infty} W e^{i\omega z/c} dz =$$

$$\Rightarrow Z_\parallel^{full} = \frac{\omega}{c}\left(-\frac{i}{c}\int_{-\infty}^{\infty} W e^{i\omega z/c} dz\right) = \frac{\omega}{c} Z$$

Impedances:

- $$Z \cong \frac{\omega}{c} Z_\parallel + x_w\left(Z_x^D x_s - \frac{1}{2} Z^Q x_w\right) + \\ + y_w\left(Z_y^D y_s + \frac{1}{2} Z^Q y_w\right)$$

Where:

- $$Z \triangleq -\frac{i}{c}\int_{-\infty}^{\infty} W e^{i\omega z/c} dz \; ; \quad Z_\parallel \triangleq \frac{1}{c}\int_{-\infty}^{\infty} W_0' e^{i\omega z/c} dz \; ;$$

$$Z^{D/Q} \triangleq -\frac{i}{c}\int_{-\infty}^{\infty} W^{D/Q} e^{i\omega z/c} dz .$$

- $$Z_\parallel^{full} = \frac{1}{c}\int_{-\infty}^{\infty} \frac{\partial W}{\partial z} e^{i\omega z/c} dz \overset{(\#)}{=} \frac{\omega}{c} \cdot Z$$

$$\Rightarrow Z_\parallel^{full} = Z_\parallel + \frac{\omega}{c} x_w\left(Z_x^D x_s - \ldots\right) + \ldots$$

$$\sim Z_\parallel^{full} \cong Z_\parallel$$

- $$\vec{Z_\perp} \triangleq \vec{\nabla}_\perp Z = \left(\frac{\partial}{\partial x_w} Z, \frac{\partial}{\partial y_w} Z\right) = (Z_x, Z_y)$$

$$\Rightarrow \vec{Z_\perp} \cong \left(Z_x^D x_s - Z^Q x_w, \; Z_y^D y_s + Z^Q y_w\right)$$

- P. W. $\rightarrow$ $$\vec{Z_\perp} = \vec{\nabla}_\perp \cdot Z = \frac{c}{\omega}\vec{\nabla}_\perp Z_\parallel^{full}$$ again, the appr... $Z_\parallel^{full}$ should be us... only when dealin... w/ the long. effec...

- Cylind. Symmetry:

$$Z_x^D = Z_y^D = Z^D, \quad Z^Q = 0$$

$$\Rightarrow \vec{Z_\perp} = Z^D (x_s, y_s) = Z^D \vec{\rho_s} = Z_\perp \vec{\rho_s}$$

- Extra: Multipole expansion approach

$$w_m(r,\theta,z) = -\frac{1}{q} I_m W_m(z) r^m \cos(m\theta), \quad \begin{cases} w_{\parallel,m} = \frac{\partial}{\partial z} w_m \\ w_{\perp,m} = \vec{\nabla}_\perp w_m \end{cases},$$

$$W_m'(z) \triangleq -\frac{\partial}{\partial z} W_m(z)$$

$$Z_{\parallel,m} \triangleq \frac{1}{c}\int_{-\infty}^{\infty} W_m'(z) e^{i\omega z/c} dz = -\frac{\omega}{c}\left(-\frac{i}{c}\int_{-\infty}^{\infty} W_m(z) e^{i\omega z/c} dz\right) = -\frac{\omega}{c} Z_{\perp,m}$$

Complement on P.W. for impedances

As we've seen, P.W. is a natural "consequence" of writing $W_{\parallel} = \frac{\partial}{\partial z} W$ and $\vec{W_{\perp}} = \vec{\nabla}_{\perp} W$. Thus, it is also incorporated when using the generalized impedance $Z = -\frac{i}{c}\int_{-\infty}^{\infty} W e^{i\omega z/c}\, dz$, together with the definitions $Z_{\parallel}^{full} = \frac{1}{c}\int_{-\infty}^{\infty} \frac{\partial}{\partial z} W e^{i\omega z/c}\, dz$ and $Z^{D/Q} = -\frac{i}{c}\int_{-\infty}^{\infty} W^{D/Q} e^{i\omega z/c}\, dz$, leading to:

$$Z_{\parallel}^{full} \cong \frac{1}{c}\int_{-\infty}^{\infty}\left[W_0' + x_w\left(W_x^{D'} x_s - \frac{1}{2} W^{Q'} x_w\right) + y_w(\dots)\right] e^{i\omega z/c}\, dz = Z_{\parallel} + x_w\left(\overbrace{\frac{1}{c}\int_{-\infty}^{\infty} W_x^{D'} e^{i\omega z/c}\, dz}^{\triangleq Z^D_{x,\parallel}} \cdot x_s - \dots\right) + \dots =$$

$$= Z_{\parallel} + x_w\left(Z^D_{x,\parallel} \cdot x_s - \frac{1}{2} Z^Q_{\parallel} \cdot x_w\right) + \dots$$

but also:

$$Z_{\parallel}^{full} \cong \frac{\omega}{c} Z = Z_{\parallel} + x_w\left(\frac{\omega}{c} Z^D_x\, x_s - \frac{1}{2}\frac{\omega}{c} Z^Q x_w\right) + \dots$$

$$\vec{Z_{\perp}} = \vec{\nabla}_{\perp} Z = \left(Z^D_x x_s - Z^Q x_w ,\ Z^D_y y_s + Z^Q y_w\right)$$

We could call these: $Z^D_{x,\perp}$, $Z^Q_{\perp}$

And thus:

$$\boxed{\begin{aligned} &\bullet\ Z^D_{x/y,\perp} = \frac{c}{\omega} Z^D_{x/y,\parallel} \\ &\bullet\ Z^Q_{\perp} = \frac{c}{\omega} Z^Q_{\parallel} \end{aligned}}$$

4. The wakes can be obtained back from the impedances by the <u>inverse transforms</u>:

$$W_0'(z) = \frac{1}{2\pi}\int_{-\infty}^{\infty} Z_{\parallel}(\omega)\, e^{-i\omega z/c}\, d\omega \;; \qquad W_t(z) = \frac{i}{2\pi}\int_{-\infty}^{\infty} Z_t(\omega)\, e^{-i\omega z/c} ,$$

where W_t refers to W_x^D, W_y^D or W^q ~ same for Z_t.

5. Reality Properties:

Since the wakes are real quantities - $W_0'^{*} = W_0'$, $W_t^{*} = W_t$ - the impedances satisfy:

$$Z_{\parallel}^{*}(\omega) = \frac{1}{c}\int_{-\infty}^{\infty} W_0'(z)\, e^{-i\omega z/c}\, dz = Z_{\parallel}(-\omega) \;\Rightarrow\; \begin{cases} \mathrm{Re}\, Z_{\parallel}(-\omega) = \mathrm{Re}\, Z_{\parallel}(\omega) & \sim \text{even} \\ \mathrm{Im}\, Z_{\parallel}(-\omega) = -\mathrm{Im}\, Z_{\parallel}(\omega) & \sim \text{odd} \end{cases}$$

$$Z_t^{*}(\omega) = +\frac{i}{c}\int_{-\infty}^{\infty} W_t(z)\, e^{-i\omega z/c}\, dz = -Z_t(-\omega) \;\Rightarrow\; \begin{cases} \mathrm{Re}\, Z_t(-\omega) = -\mathrm{Re}\, Z_t(\omega) & \sim \text{odd} \\ \mathrm{Im}\, Z_t(-\omega) = \mathrm{Im}\, Z_t(\omega) & \sim \text{even} \end{cases}$$

6. Passivity and Positivity

Using the harmonic decomposition of the beam current, the total energy lost by the beam to the chamber can be written as:

$$U_{loss} = \frac{1}{\pi}\int_{0}^{\infty} |\hat{I}(\omega)|^2 \cdot \mathrm{Re}\{Z_{\parallel}(\omega)\}\, d\omega , \qquad \text{where clearly } |\hat{I}(\omega)|^2 \geq 0 .$$

Thus, for a passive chamber: $U_{loss} \geq 0$ for any current distribution

$\therefore$ Since we can choose $\hat{I}(\omega)$ concentrated around any specific frequency ω $\;\longrightarrow\;$ $\mathrm{Re}\{Z_{\parallel}(\omega)\} \geq 0 \quad \forall \; \omega \geq 0$

And since $\mathrm{Re}\{Z_{\parallel}(-\omega)\} = \mathrm{Re}\{Z_{\parallel}(\omega)\}$, this means:

$$\mathrm{Re}\{Z_{\parallel}(\omega)\} \geq 0 \quad \text{for all } \omega$$

Now, for the transverse impedances, something similar can be derived for the dipolar terms:

$\rightarrow$ For a displaced beam – described by a dipole-current moment $D(\omega) \sim x_s \hat{I}(\omega)$ – the contribution of the dipole channel to the energy loss can be written:

$$U_D \propto \int_0^{\infty} |D(\omega)|^2 \cdot \mathrm{Re}\{Z_{\parallel,D}(\omega)\}\, d\omega \quad ; \quad \text{where} \quad Z_{\parallel,D} = \frac{1}{c}\int_{-\infty}^{\infty} W^{D'}_{x/y}(z)\, e^{i\omega z/c}\, dz$$

(remembering that $W_{\parallel}^{full}(z) = \frac{\partial}{\partial z} W(z) = W_0'(z) + x_w W_x^{D'} x_s + y_w W_y^{D'} y_s + \dots$)

Integrating by parts:

$$\frac{1}{c}\int_{-\infty}^{\infty} W^{D'}_{x/y}\, e^{i\omega z/c}\, dz = \frac{\omega}{c}\left(-\frac{i}{c}\int_{-\infty}^{\infty} W^{D}_{x/y}\, e^{i\omega z/c}\, dz\right) = \frac{\omega}{c} Z^{D}_{x/y}$$

And thus, passivity implies:

$$U_D \geq 0 \text{ for any } D(\omega) \Rightarrow \mathrm{Re}\{Z_{\parallel,D}(\omega)\} \geq 0 \Rightarrow \frac{\omega}{c}\mathrm{Re}\{Z^{D}_{x/y}\} \geq 0 \quad \forall\, \omega \geq 0$$

$\therefore$ Since $\mathrm{Re}\{Z_t(-\omega)\} = -\mathrm{Re}\{Z_t(\omega)\}$: $\Rightarrow \mathrm{Re}\{Z^{D}_{x/y}\} \geq 0 \quad \forall\, \omega \geq 0$

$$\begin{cases} \mathrm{Re}\{Z^{D}_{x/y}(\omega)\} \geq 0, & \omega \geq 0 \\ \mathrm{Re}\{Z^{D}_{x/y}(\omega)\} \leq 0, & \omega < 0 \end{cases}$$

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* For the quadrupolar impedance, passivity <u>does not imply</u> $\mathrm{Re}\{Z^{Q}(\omega)\} \geq 0$ because, while $U_D \propto |D|^2 \frac{\omega}{c}\mathrm{Re}\{Z^{D}_{x/y}(\omega)\}$ (which $\Rightarrow \omega \cdot \mathrm{Re}\{Z^{D}\} \geq 0$), $U_Q \propto \mathrm{Re}\{Z^{Q} \overset{\text{monopole moment}}{I_0} \underset{\text{quad. moment}}{Q_b^*}\}\cdot\frac{\omega}{c}$, and $I_0 Q_b^*$ can have any sign

7. Causality and Kramers - Kronig

Due to the causality of the wakes - $W(z<0) = 0$ -, the real and imaginary parts of the impedances cannot be completely arbitrary: they satisfy the so-called Kramers - Kronig relations, i.e., are connected by a Hilbert transform.

To derive and outline such connexion, let's write $Z_{\parallel}(\omega) = R_{\parallel}(\omega) + i X_{\parallel}(\omega)$.

Then, since the wakes are real:

$$\begin{cases} R_{\parallel}(\omega) = Re\{Z_{\parallel}(\omega)\} \Rightarrow R_{\parallel}(-\omega) = R_{\parallel}(\omega) \\ X_{\parallel}(\omega) = Im\{Z_{\parallel}(\omega)\} \Rightarrow X_{\parallel}(-\omega) = -X_{\parallel}(\omega) \end{cases}$$

From the inverse transform:

$$W_{\parallel}(z) = \frac{1}{2\pi}\int_{-\infty}^{\infty} Z_{\parallel}(\omega)\, e^{-i\omega z/c}\, d\omega = \frac{1}{2\pi}\left[\int_{-\infty}^{0} Z_{\parallel}(\omega)\, e^{-i\omega z/c}\, d\omega + \int_{0}^{\infty} Z_{\parallel}(\omega)\, e^{-i\omega z/c}\, d\omega\right] =$$

$$= \frac{1}{2\pi}\left[+\int_{0}^{\infty}\left(R_{\parallel}(-\omega) + i X_{\parallel}(-\omega)\right) e^{-i\omega z/c}\, d\omega + \int_{0}^{\infty}\left(R_{\parallel}(\omega) + i X_{\parallel}(\omega)\right) e^{i\omega z/c}\, d\omega\right] =$$

$$= \frac{1}{2\pi}\int_{0}^{\infty}\left[R_{\parallel}(\omega)\cdot\overbrace{\left(e^{i\omega z/c} + e^{-i\omega z/c}\right)}^{2\cdot\cos(\omega z/c)} + i X_{\parallel}(\omega)\cdot\overbrace{\left(e^{i\omega z/c} - e^{-i\omega z/c}\right)}^{-2i\cdot\sin(\omega z/c)}\right] d\omega =$$

$$= \frac{1}{\pi}\int_{0}^{\infty}\left[R_{\parallel}(\omega)\cos(\omega z/c) + X_{\parallel}(\omega)\sin(\omega z/c)\right] d\omega\,.$$

Now, to make use of causality, let $u > 0$ and evaluate $W_{\parallel}(z = -u)$:

$$W_{\parallel}(-u) = \frac{1}{\pi}\int_0^{\infty}\left[R_{\parallel}(\omega)\cos(\omega u/c) - X_{\parallel}(\omega)\sin(\omega u/c)\right]d\omega = 0$$

$$\Rightarrow \int_0^{\infty} R_{\parallel}(\omega)\cos\left(\frac{\omega u}{c}\right)d\omega = \int_0^{\infty} X_{\parallel}\sin\left(\frac{\omega u}{c}\right)d\omega$$

Then, for $z > 0$, we can write (from (I)):

$$W_{\parallel}(z) = \frac{2}{\pi}\int_0^{\infty} R_{\parallel}(\omega)\cos\left(\frac{\omega z}{c}\right)d\omega \quad \text{(II)} \qquad \text{or} \qquad W(z) = \frac{2}{\pi}\int_0^{\infty} X_{\parallel}(\omega)\sin\left(\frac{\omega z}{c}\right)d\omega \quad \text{(III)}$$

$\Rightarrow$ * The knowledge of either $Re\{Z_{\parallel}\}$ or $Im\{Z_{\parallel}\}$ alone is enough to reconstruct the wake $W_{\parallel}$

Now, to derive Kramers-Kronig relations we see that:

$$Z_{\parallel}(\omega) = \frac{1}{c}\int_{-\infty}^{\infty} W_{\parallel}(z)\, e^{i\omega z/c}\,dz = \frac{1}{c}\int_0^{\infty}\left[W_{\parallel}\cos\left(\frac{\omega z}{c}\right) + i W_{\parallel}\sin\left(\frac{\omega z}{c}\right)\right]dz = R_{\parallel}(\omega) + i X_{\parallel}(\omega)$$

$\sin(a+b) = \sin a\cos b + \sin b\cos a$
$\sin(a-b) = \sin a\cos b - \sin b\cos a$
$\Rightarrow 2\sin a\cos b = \sin(a+b) + \sin(a-b)$

$$\Rightarrow R_{\parallel}(\omega) = \frac{1}{c}\int_0^{\infty} W_{\parallel}(z)\cos\left(\frac{\omega z}{c}\right)dz \overset{\text{(III)}}{=} \frac{2}{\pi c}\int_0^{\infty}\int_0^{\infty} X_{\parallel}(\omega')\sin\left(\frac{\omega' z}{c}\right)\cos\left(\frac{\omega z}{c}\right)d\omega'\,dz =$$

$$= \frac{\cancel{2}}{\pi c}\int_0^{\infty}\left(\int_0^{\infty}\frac{1}{\cancel{2}}\left[\sin\left(\frac{(\omega'+\omega)z}{c}\right) + \sin\left(\frac{(\omega'-\omega)z}{c}\right)\right]dz\right)d\omega$$

Using the strategy of introducing a small damping factor $e^{-\eta z}$ $(\eta > 0)$

$$\int_0^\infty \sin\left(\frac{az}{c}\right) dz = \lim_{\eta \to 0^+} \int_0^\infty e^{-\eta z} \sin\left(\frac{az}{c}\right) dz = \lim_{\eta \to 0^+} \frac{ac}{a^2 + \eta^2 c^2} = \frac{c}{a}$$

However, if "a" can pass through zero inside another, like in our case of $a = \omega' - \omega$, this result must be interpreted in the Cauchy principal value sense: $c \cdot P \frac{1}{a}$.

$$\frac{1}{\omega' + \omega} + \frac{1}{\omega' - \omega} = \frac{2\omega'}{\omega'^2 - \omega^2}$$

Thus:

$$R_{11}(\omega) = \frac{1}{\pi c} \int_0^\infty \left(\frac{c}{\omega' + \omega} + P \frac{c}{\omega' - \omega} \right) X_{11}(\omega') d\omega' = \frac{2}{\pi} P \int_0^\infty \frac{\omega' X_{11}(\omega')}{\omega'^2 - \omega^2} d\omega'$$

And analogously:

$$X_{11}(\omega) = \frac{1}{c} \int_0^\infty W_{11}(z) \sin\left(\frac{\omega z}{c}\right) dz \overset{(\mathrm{II}) + \text{algebra}}{\Longrightarrow} X_{11}(\omega) = -\frac{2\omega}{\pi} P \int_0^\infty \frac{R_{11}(\omega')}{\omega'^2 - \omega^2} d\omega'$$

Or equivalently, opening $\frac{2\omega'}{\omega'^2 - \omega^2}$ back into $\frac{1}{\omega' + \omega} + \frac{1}{\omega' - \omega}$ and manipulating to express these relations in terms of the whole frequency axis, gives the standard Kramers - Kronig relations:

$$R_{11}(\omega) = \frac{1}{\pi} P \int_{-\infty}^\infty \frac{X_{11}(\omega')}{\omega' - \omega} d\omega' \; ; \qquad X_{11}(\omega) = -\frac{1}{\pi} P \int_{-\infty}^\infty \frac{R_{11}(\omega')}{\omega' - \omega} d\omega'$$

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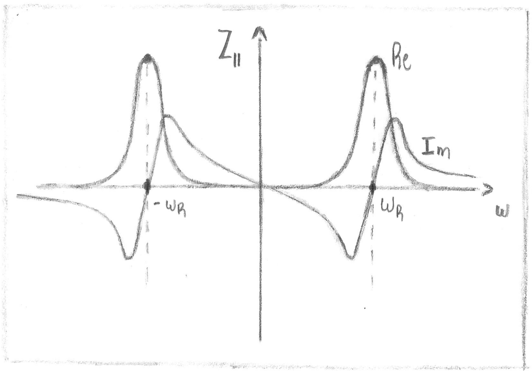
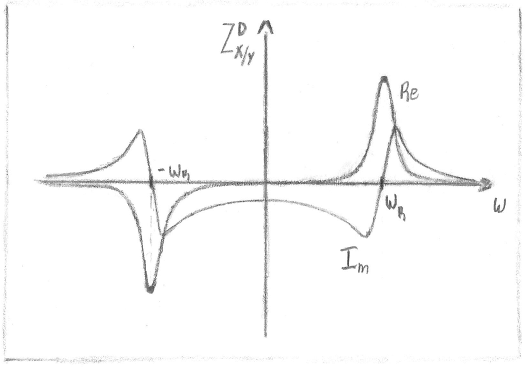
Proceeding analogously for the transverse impedances $\left(Z_t(\omega) = R_t(\omega) + i\, X_t(\omega) \right)$ also gives:

$$R_t(\omega) = \frac{1}{\pi} P \int_{-\infty}^{\infty} \frac{X_t(\omega')}{\omega' - \omega}\, d\omega' \qquad X_t(\omega) = -\frac{1}{\pi} P \int_{-\infty}^{\infty} \frac{R_t(\omega')}{\omega' - \omega}\, d\omega'$$

~ Kromers - Kronig relations for the transverse impedances

To illustrate many of the presented properties of the real and imaginary arts of $Z_{\parallel}(\omega)$ and $Z_t(\omega)$, we can draw these curves for the case of a resonator odel - one of the models that will be explored in the next section - with resonant equency ω_R:





Adapted from [S. Khan]